

BUZABRICI-FILIPESCU ROBERT

Senior Embedded Software Engineer

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Location: Iasi, Romania | Languages: • RO Native • EN C2 • DE A2

Highly accomplished **Senior Embedded Software Engineer** with over 13 years of experience in automotive and space-grade systems. Specialized in instrument cluster development, CAN communication, and AUTOSAR-based platforms. Proven track record of delivering robust, high-performance embedded solutions for leading OEMs including Daimler, Ford, Renault, Volkswagen, Porsche, and Audi. Strong focus on reliability, real-time performance, and technical leadership. Proficient in C, AUTOSAR, CANoe, Trace32, V-Cycle, and Agile/SCRUM methodologies. Experienced in ASPICE compliance, functional safety (ISO 26262), and stakeholder alignment. Currently expanding expertise in C++, C#, Python, and Rust to drive innovation. Seeking challenging opportunities in collaborative, remote-friendly environments that value technical excellence, mentorship, and continuous improvement.

Key Achievements

- **Debugging Efficiency:** Reduced debugging time by 25% through rigorous code reviews and adherence to coding standards.
- **Code Reliability:** Enhanced code reliability by 25% via comprehensive unit and component testing strategies (e.g., Google Test, VectorCast).
- **Consistent Contributions:** Delivered over 300 high-quality Git commits, ensuring consistent quality and timely releases.
- **Knowledge Management:** Built a centralized knowledge base of 50+ software modules, cutting integration errors by 20% and boosting cross-team collaboration.
- **Performance Optimization:** Optimized embedded modules, achieving 20% memory reduction and improved responsiveness for smoother OEM integration.
- **Leadership:** Acted as Tech Lead, resolving critical roadblocks, mentoring new members, and ensuring project continuity with direct client engagement.

Technical Skills

- **Programming Languages:** Embedded C, ANSI C, C++, C# (Intermediate), Python (Intermediate), Rust (Exploring)
- **Automotive & Space Systems:** Classic & Adaptive AUTOSAR (Vector Certified Embedded Associate), EB Tressos, CAN communication, OSEK RTOS, Powertrain systems, Inter-process communication
- **Tools & Technologies:** CANoe, CANalyzer, Trace32, Google Test, Git, Jira, Confluence, Microsoft Visual Studio Code, Jenkins, Xilinx SDK, Zynq-7000 SoC, ARM Cortex-A9, VectorCast, PVCS, MKS, IMS, RTC (Rational Team Concert), Multi IDE, Serena TeamTrack, INCA, Linux, CAPL scripting, Visual C++ 6.0 IDE, Renesas flashing tools
- **Protocols & Standards:** CAN, CCP, XCP, ISO 26262 (ASIL), ASPICE, SpaceWire driver, Franca IDL (.fidl)
- **Methodologies:** V-Cycle, Agile/SCRUM, Functional & System Architecture (UML, MagicDraw), Requirements Traceability, Functional Safety

Experience

Capgemini Engineering | Iasi, Romania

Senior Embedded Software Engineer / Tech Lead Replacement | April 2022 – November 2025

- Optimized embedded modules, achieving 20% memory reduction and improved system responsiveness for smoother OEM integration.
- Boosted cross-functional collaboration, cutting issue resolution time by 30%.
- Enhanced test coverage by refining CAPL scripts and test campaigns, reducing post-release bugs by 30%.
- Delivered 300+ Git commits with 100% coding standards compliance, leading to 25% less debugging time.
- Acted as Tech Lead, resolving critical roadblocks, mentoring new members, and maintaining project continuity with direct client engagement.

ENEA Software Development Services | Iasi, Romania

Senior Embedded Software Engineer – Space Platforms | April 2021 – April 2022

- Improved real-time performance by reducing latency via buffer management and interrupt-driven design.
- Optimized constrained environments, working with Xilinx SDK and Zynq-7000 SoC (ARM Cortex-A9).
- Developed space-grade C/C++ modules with a focus on reliability and fault tolerance.
- Engineered SpaceWire driver interface and state machine for RendezVous Sensor, enabling real-time telecommand processing.
- Championed a knowledge base of 30+ modules, cutting integration errors by 20% and streamlining communication across 3 teams.

Visteon Electronics EOOD | Sofia, Bulgaria (Contractor from RINF.TECH)

Senior Embedded Software Engineer – Automotive Platforms | June 2014 – May 2020

- Led instrument cluster development for Daimler (C/C++) projects (VS Daimler 2.0, MFA2, MRA2), improving feature reliability and user experience.
- Validated display components with CANoe, cutting test cycle time by 20% and delivering robust OEM features (Speedometer, Tachometer, Odometer, etc.).
- Implemented IPC & Franca IDL files, enabling seamless communication between Application Layer & HMI, and ensuring AUTOSAR RTE compliance.

Continental Automotive Romania SRL | Iasi, Romania

Embedded Software Engineer | August 2010 – May 2014

- Implemented customer-specific C/C++ functionalities across Ford, Renault, VW, Porsche, Audi platforms.
- Improved diagnostic reliability across 5+ OEM platforms by configuring faults (DTC), freeze frames, and failure reactions.
- Validated CAN channels and diagnostic instances, enhancing error management.
- Reduced field failures via thorough unit & integration testing and documented Q&A reports.
- Applied ISO 26262 standards in Powertrain systems.

Education

- Master's Degree, Telecommunication Networks | (2009 - 2011)
Faculty of Electronics, Telecommunications and Information Technology, Iași, Romania
- Bachelor's Degree, Information Technology | (2005 - 2009)
Faculty of Automatic Control and Computer Engineering, "Gh.Asachi" University, Iași, Romania
- Bachelor's Degree, Economics and Management | (2007 - 2010)
Faculty of Economics and Business Administration, "Al. I. Cuza" University, Iasi, Romania

Certifications

Capgemini L2 Automotive Industry Certification • Vector Certified Embedded Associate (Vector) • The Bits and Bytes of Computer Networking (Coursera) • Responsive Web Design (freeCodeCamp) • Technical Support Fundamentals (Coursera)